

FILTER DESIGN TEST CASE

High-Pass Filter - Band 25 (1900 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting Band 25 (1900 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on Langasite substrate with a target center frequency of 4779.6 MHz and bandwidth of 208.6 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: Band 25 (1900 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: Langasite
- Package: Flip-Chip BGA
- Design Tool: PathWave EM Design
- Design Center: Osaka Design Center

This specification applies to all variants of the High-Pass Filter design targeting Band 25 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	4779.6 MHz
Bandwidth (3 dB):	208.6 MHz
Insertion Loss Target:	≤ 3.3 dB
Return Loss Target:	≥ 11.0 dB
Out-of-Band Rejection:	≥ 51.0 dB
Isolation (Duplexer):	≥ 28.0 dB
Q Factor:	6587
Temperature Coefficient:	-19.6 ppm/°C
Die Size:	2.8 x 0.9 mm
Wafer Size:	8-inch
Number of Resonators:	4
Electrode Material:	Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open PathWave EM Design and load the High-Pass Filter template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 4779.6 MHz, BW 208.6 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, Langasite).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 14339 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.3 dB
- Return loss in passband shall be >= 11.0 dB
- Out-of-band rejection at +/- 313 MHz offset >= 51.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 11.7 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2024-11-27
Devices Tested: 46
Insertion Loss (meas): 3.04 dB
Return Loss (meas): 12.7 dB
Rejection (meas): 54.1 dB
Isolation (meas): 33.0 dB
Q Factor (meas): 6587
Group Delay Variation: 3.9 ns
Power Handling: +31 dBm
Die Yield: 87.9%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The High-Pass Filter design demonstrated excellent performance for Band 25 (1900 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author:	Dr. Raj Patel	Date:	2024-11-27
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Approver:	Dr. James Liu	Date:	2024-12-07

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